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E. Fast Series Drift

time limit per test: 2.5 seconds

memory limit per test: 256 megabytes

Forbidden Keywords for the Quiz: `open`, `file`, `creat`, `fstream`, `thread`, `process`, `system`, `exec`

You are given three integers a , n and m . Calculate $(a^1 + a^2 + \dots + a^n) \% m$.

Input

The first line contains an integer T ($1 \leq T \leq 10^5$) — total numbers of test cases.

In each of the next T test cases, there are three integers a ($1 \leq a \leq 10^6$), n ($1 \leq n \leq 10^{12}$) and m ($1 \leq m \leq 10^9$)

Output

Print one integer — the result of $(a^1 + a^2 + \dots + a^n) \% m$.

Example

input	Copy
3 2 5 1000 2 9 1000 1 100 30	
output	Copy
62 22 10	

CSE221 Assignment 03 Summer 2026

Contest is running

2 days

Contestant



→ Submit?

Language: Java 21 64bit Choose file: Choose File No file chosen

Success!



Submit

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